

HR Attrition Analysis – EDA Report

1. Project Overview

Objective

The objective of this project is to analyze employee attrition data and identify the factors that contribute to employee turnover. The insights generated from this analysis can help organizations improve employee retention strategies.

Dataset

The dataset contains employee demographic, job-related, and satisfaction-related information along with attrition status.

2. Data Understanding

Dataset Shape

- Number of Rows: 1470
- Number of Columns: 35

Data Types

The dataset contains:

- Numerical Features
- Categorical Features

Target Variable

- Attrition (Yes/No)
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3. Data Cleaning

The following preprocessing steps were performed:

- Checked dataset structure using `info()` and `shape`.
- Identified numerical and categorical columns.
- Checked for missing values.
- Removed duplicate records.
- Cleaned categorical values where necessary.
- Verified data quality before analysis.

4. Feature Engineering

The following new features were created:

Salary Category

Employees were categorized into:

- Low Salary
- Medium Salary
- High Salary

Experience Level

Employees were grouped into:

- Junior
- Mid-Level
- Senior

These engineered features helped improve business understanding of attrition patterns.

5. Exploratory Data Analysis

Univariate Analysis

Performed analysis on:

- Age Distribution
- Monthly Income Distribution
- Department Distribution
- Attrition Distribution

Key Observation:

Employee attrition is unevenly distributed across the workforce.

Bivariate Analysis

Analyzed relationships between:

- Attrition vs Overtime
- Attrition vs Department

- Attrition vs Monthly Income
- Attrition vs Job Satisfaction

Key Observation:

Employees working overtime and employees with lower income levels show higher attrition rates.

Multivariate Analysis

Analyzed multiple factors simultaneously to identify complex attrition patterns.

Key Observation:

Attrition is influenced by a combination of salary, experience level, satisfaction, and overtime.

6. Correlation Analysis

A correlation heatmap was generated to identify relationships between numerical variables.

Key Findings:

- Strong positive correlations exist among several employee experience-related variables.
- Some variables show moderate relationships with attrition-related factors.

7. Business Insights

1. Employees with lower salaries are more likely to leave the organization.
2. Overtime significantly increases attrition risk.
3. Low job satisfaction is associated with higher employee turnover.
4. Early-career employees exhibit higher attrition rates.
5. Certain departments experience higher attrition than others.
6. Employee retention can be improved through workload management.
7. Compensation and career growth opportunities play an important role in retention.

8. Feature Engineering Ideas

1. Overtime Risk Score

2. Satisfaction Index
3. Promotion Delay Indicator
4. Department Attrition Risk
5. Employee Engagement Score

These features can be useful for future predictive modeling projects.

9. Recommendations

- Reduce excessive overtime.
 - Improve employee engagement programs.
 - Review compensation structures.
 - Provide better career growth opportunities.
 - Conduct regular employee satisfaction surveys.
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10. Conclusion

This analysis identified salary, overtime, job satisfaction, and experience level as key drivers of employee attrition. Employees with low satisfaction, lower salaries, and overtime responsibilities are at greater risk of leaving the organization. The findings can support HR teams in developing targeted retention strategies and improving workforce stability.